Internship Assignment: Mini Construction Management System

1. Assignment Overview

Develop a small **Construction Management System** that allows a construction company to create projects, define their BOQ, manage materials, and record basic project progress.

Duration: August 18–31, 2026
Duration: 2 weeks

The goal is to demonstrate practical full-stack development using the assigned technology stack.

2. Technology Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- TanStack Query
- TanStack Table
- React Hook Form
- Zod

Backend

- NestJS
- REST API
- Swagger/OpenAPI

Database

- PostgreSQL
- Prisma ORM

Development

- Docker / Docker Compose
- Git / GitHub
- ESLint
- Prettier

3. Scope

The system will contain **four core areas**:

1. Projects
2. BOQ
3. Materials & Inventory
4. Project Progress

No full accounting, procurement workflow, HR, payroll, subcontractor management, or mobile application is required.

4. Project Management

Users should be able to:

- Create a project
- View projects
- Edit a project
- Delete a project
- View project details

Project fields

- Project name
- Project code
- Client name
- Location
- Start date
- End date
- Budget
- Status

Status

- Planned
- Ongoing
- Completed

5. BOQ Management

Each project should have a simple **Bill of Quantities**.

BOQ fields

- Description

- Unit
- Quantity
- Unit price
- Total

Automatically calculate:

Total = Quantity × Unit Price

Display:

Total BOQ Value

The intern should implement:

- Add BOQ item
- Edit BOQ item
- Delete BOQ item
- View project BOQ

6. Materials & Inventory

Create a simple material inventory.

Material

- Material name
- Code
- Unit
- Current stock
- Minimum stock

Example:

- Cement
- Steel
- Sand
- Gravel
- Brick

Inventory operations

Only two operations are required:

Stock In

- Material
- Quantity
- Date

- Reference

Stock Out

- Material
- Project
- Quantity
- Date
- Reference

Business Rule

The system must not allow:

Stock Out > Available Stock

Display a warning when stock reaches the minimum level.

7. Project Progress

The Project Manager should be able to record basic progress.

Progress Record

- Project
- Date
- Description
- Progress %
- Notes

Example:

Foundation work completed — 35%

The project page should display the latest progress percentage.

8. Dashboard

Create a simple dashboard showing:

Projects

- Total projects
- Ongoing projects
- Completed projects

Inventory

- Total materials
- Low-stock materials

Project Performance

For each project:

- Project name
- Budget
- BOQ value
- Progress %
- Status

9. Frontend Requirements

Forms

Use:

React Hook Form + Zod

for:

- Project form
- BOQ form
- Material form
- Stock transaction form
- Progress form

Data Fetching

Use **TanStack Query** for API communication.

Tables

Use **TanStack Table** for:

- Projects
- BOQ
- Materials
- Inventory transactions

Implement basic:

- Search
- Sorting
- Pagination

10. Backend Requirements

Build REST APIs using NestJS.

Required modules:

- Projects
- BOQ
- Materials
- Inventory
- Progress

Use:

- Controllers
- Services
- DTOs
- Validation
- Proper error handling

Document the API using **Swagger**.

11. Database

Use PostgreSQL with Prisma.

Minimum entities:

```
projects
boq_items
materials
inventory_transactions
progress_records
```

Implement proper relationships and Prisma migrations.

12. Docker

The application should run using Docker Compose.

At minimum:

- Next.js
- NestJS
- PostgreSQL

The README should explain how to:

- Start the system
- Run migrations
- Seed sample data
- Stop the system

13. Testing

Focus testing on the most important business rules:

- Create project
- Calculate BOQ total
- Stock In
- Stock Out
- Prevent negative stock
- Record project progress

14. Two-Week Plan

Week 1

August 18–19

Project Setup + Database

- Git repository
- Docker
- PostgreSQL
- Prisma
- NestJS
- Next.js
- Database schema

August 20–22

Projects + BOQ

- Project CRUD
- Project details
- BOQ CRUD
- BOQ calculations

August 23–24

Materials + Inventory

- Material management
- Stock In

- Stock Out
- Stock balance

Week 2

August 25–26

Project Progress

- Progress records
- Progress percentage
- Project progress view

August 27–28

Dashboard + UI

- Dashboard
- Tables
- Search
- Filtering
- Form validation

August 29

Testing & Documentation

- Tests
- Bug fixing
- Swagger
- README
- Seed data

August 30–31

Finalization & Demonstration

- Final testing
- Docker verification
- Code cleanup
- Final presentation
- Demonstration

Final Deliverables

- Working web application
- NestJS API
- PostgreSQL database

- Prisma schema/migrations
- Swagger documentation
- Docker Compose
- Basic automated tests
- GitHub repository
- README
- Final demonstration